WORKED SOLUTION

# STEP III 1989, Question 14

Mechanics • Pulleys and Impulse

**STEP III 1989, Question 14**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q14

## The key idea

For part (i), cut the disc into thin rings: each ring has all its mass at one distance from the centre, so both the mass and the moment of inertia are single integrals in $r$. For part (ii), the whole system has one degree of freedom (the depth of $D$), so one equation of motion gives the acceleration. When $B$ is suddenly stopped, the string constraint tells us at once how fast $D$ can then move, and the impulse is just the change in its momentum.

## Solution

### Part (i)

(The axis in part (i) is the one through the centre $O$; the "$C$" in the paper is a misprint.) Let the density be $\rho = kr$ at distance $r$ from $O$, with $k$ constant. The thin ring between radii $r$ and $r + \mathrm{d}r$ has area $2\pi r\,\mathrm{d}r$, so mass $2\pi kr^2\,\mathrm{d}r$, and every point of it is at distance $r$ from the axis. Hence

$$m = \int_0^a 2\pi kr^2\,\mathrm{d}r = \frac{2\pi ka^3}{3}, \qquad I = \int_0^a r^2 \cdot 2\pi kr^2\,\mathrm{d}r = \frac{2\pi ka^5}{5}.$$

Dividing, $I/m = \frac{3}{5}a^2$, so

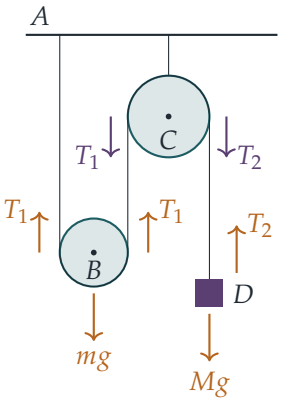
$$I = \tfrac{3}{5}ma^2,$$

as required. (It is bigger than the uniform disc's $\frac{1}{2}ma^2$, as it should be: more of the mass is near the rim.)

### Part (ii): the motion before $B$ is stopped

Model: $B$ is a smooth pulley of mass $m$, and $C$ is a disc of the type in (i) with mass $m$ and radius $a$, so its moment of inertia about its fixed axis is $\frac{3}{5}ma^2$. The string is light and inextensible and does not slip on $C$. Because $B$ is smooth, the tension $T_1$ is the

same in the two strands $AB$ and $BC$. Call the tension in the strand $CD$ $T_2$; since $C$ is rough, $T_2$ need not equal $T_1$.




Let $D$ move down with acceleration $f$. The two strands supporting $B$ each shorten by half of the amount by which $CD$ lengthens, so $B$ rises with acceleration $\frac{1}{2}f$; and since the string does not slip on $C$, the angular acceleration of $C$ is $f/a$. The equations of motion are:

$$\begin{aligned} D\ (\text{down}): &\quad Mg - T_2 = Mf, \\ B\ (\text{up}): &\quad 2T_1 - mg = \tfrac{1}{2}mf, \\ C\ (\text{moments about its axis}): &\quad (T_2 - T_1)a = \tfrac{3}{5}ma^2 \cdot \frac{f}{a}. \end{aligned}$$

Substituting $T_2 = Mg - Mf$ and $T_1 = \frac{1}{2}mg + \frac{1}{4}mf$ into the last equation,

$$Mg - Mf - \tfrac{1}{2}mg - \tfrac{1}{4}mf = \tfrac{3}{5}mf \quad \Longrightarrow \quad f\left(M + \tfrac{17}{20}m\right) = g\left(M - \tfrac{1}{2}m\right),$$

so

$$f = \frac{10(2M - m)g}{20M + 17m}.$$

This is constant, and positive precisely because $m < 2M$: that is why the condition is there. It tells us $D$ really does descend from rest. So after $t$ seconds $D$ is moving down with speed

$$v = ft = \frac{10(2M - m)gt}{20M + 17m},$$

$B$ is rising at $\frac{1}{2}v$ and $C$ is turning with angular speed $v/a$.

(An energy check gives the same $f$: the kinetic energy is $\frac{1}{2}(M + \frac{1}{4}m + \frac{3}{5}m)\dot{x}^2$ when $D$ has fallen $x$, and the potential energy lost is $(M - \frac{1}{2}m)gx$.)

**Part (ii): the impulse**

**Ask yourself** After $B$ is stopped, can $D$ still move?

While the string is taut, the speed of $D$ is always twice the speed of $B$, because $A$ is fixed and $AB$ and $BC$ each change length by the same amount. Once $B$ is at rest, both of these lengths are fixed, so no string can pass round $C$ and $D$ must stop too, as must $C$. The string does stay taut: $D$ is moving away from $C$, so it pulls $CD$ tight, and $C$, still turning, draws string up from the $BC$ side, so that strand is pulled tight as well. (In fact the impulsive tensions come out as $Mv$ in $CD$ and $(M + \frac{3}{5}m)v$ in $BC$, both positive, which confirms this.)

So the velocity of $D$ drops from $v$ downwards to zero, and the impulse on $D$ (from the string) equals its change of momentum:

$$\text{impulse on } D = Mv = \frac{10M(2M - m)gt}{20M + 17m} \quad \text{vertically upwards.}$$

## Checks

- If $m \to 0$ (light pulleys) then $f \to g$ and the impulse is $Mgt$: $D$ simply falls freely, as expected.
- If $m = 2M$ the impulse is zero: the system is in equilibrium and never starts moving.
- With $M = 3$, $m = 2.5$, $g = 9.8$, $t = 1.3$, stepping the motion numerically gives $Mv \approx 13.0507$, matching the formula, and the kinetic energy gained matches the potential energy lost. The moment of inertia ratio from numerical integration is 0.6000.

## Common slips

- Taking the same tension on both sides of $C$. The pulley $C$ has mass and is rough, so the tension difference is exactly what turns it.
- Forgetting that $B$ moves at half the speed of $D$, which gives $\frac{1}{4}m$, not $m$, in the effective mass.
- Giving the impulse without its direction: it acts upwards on $D$.

## Beyond STEP

Writing the kinetic energy as $\frac{1}{2}M_{\text{eff}}\dot{x}^2$ in a single coordinate is the starting point of Lagrangian mechanics, where constraints like the pulley geometry are built in from the beginning and the tensions never appear. Sudden constraints, like stopping $B$, are handled the same way with impulsive equations. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body rotation and impulse in this style.

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